01 · OMS architecture: order capture, routing & orchestration

Supply Chain Reference Architecture v2.2 · mrsupplychain.ai

1 WHAT IT IS

An Order Management System (OMS) captures orders from every channel, checks availability, decides which node fulfills each order — DC, store or supplier — and orchestrates execution with the WMS and TMS while keeping the customer informed.

2 WHY IT MATTERS

When each channel runs its own order logic, the same unit is sold twice, orders ship from the wrong node and nobody has one view of order status. **The OMS is the decision layer of omnichannel fulfillment: it decides how each order is served, while the WMS and TMS execute.** Routing rules are business decisions, so commercial and supply chain teams must design them together.

3 HOW IT IS DONE

1. **Unify order capture.** All channels feed one order record with one status model.
2. **Give the OMS a single inventory view.** Every node, in near real time, with reservations.
3. **Write routing rules with the business.** Cost, promise, customer priority and node capacity, in explicit order.
4. **Integrate execution and measure routing.** Real-time links to WMS and TMS; track how often orders go to the optimal node.

IN PRACTICE

The OMS is the most important system in the omnichannel architecture — not the WMS or TMS

The WMS and TMS are execution systems. The OMS is the decision system that defines how each order is served. Without an OMS that orchestrates correctly, the most sophisticated execution systems cannot deliver the omnichannel experience customers expect.

Real-time OMS-WMS-TMS integration is the prerequisite of promise accuracy

An OMS that makes a delivery promise without real-time confirmation from the WMS (is inventory available?) and the TMS (how long will transit take?) will generate promises that cannot be kept.

Define OMS routing rules with the commercial team — not just logistics

OMS routing rules have commercial implications (prioritize strategic customer) and logistical implications (nearest node, lowest cost). Rule design requires input from both teams.

Measure order routing accuracy — the % of orders routed to the optimal node

It's the metric that verifies whether the OMS routing engine is functioning correctly and whether the business rules reflect the company's current priorities.

ILLUSTRATIVE CASE

Illustrative case built from typical industry values — not data from a specific company.

> **Illustrative case: Omnichannel OMS implementation — retailer with 5 channels and 4 inventory nodes** — Retailer has 5 sales channels (web, app, store, call center, marketplace) and 4 nodes (national DC, regional DC, online stores, dropship). Without an OMS, each channel has its own routing logic and inventory management.

OMS functionality	Before (no OMS)	After (with OMS)
Unified inventory visibility across 4 nodes	HIGH	Siloed inventory by channel · Frequent overselling when same item sold in 2 channels
Automatic routing to lowest-cost/lead-time node	HIGH	Manual, ad-hoc rules · 28% of orders suboptimally routed
Unified delivery promise across all channels	MEDIUM	Generic lead time by channel · Not differentiated by node or destination
Real-time order status visibility for customer and CS	HIGH	0% real-time visibility · CS managed updates by email

Result: Post-OMS: Overselling eliminated. Optimal routing: 94% of orders to the correct node. Promise accuracy: 91.4% -> 97.2%. CS query resolution time: 8.4 min -> 2.1 min with real-time order status visibility.

HOW IT IS MEASURED

Order Routing Accuracy % (orders routed to the optimal node)

Order Routing Accuracy %(Orders routed to the lowest-cost + lead-time node per OMS policy / Total orders routed) × 100

Benchmark: >95% of orders routed to the optimal node by the OMS engine Incorrect routing creates two simultaneous problems: the order is served from a suboptimal node (higher cost or longer lead time) and the inventory at the optimal node remains unused.

Order Cycle Time (time from order capture to delivery)

Order Cycle TimeConfirmed delivery date/time - Order capture date/time

Benchmark: <24 hours for same-day e-commerce · <48 hours for next-day · <5 business days for standard B2B · Varies by channel promise Order cycle time is the KPI that measures the total speed of the end-to-end logistics chain. Differentiation by channel (same-day, next-day, standard) reflects the different value propositions to the customer.

WHAT YOU WOULD USE

OMS Platforms

Manhattan Active Omni — Module: Omnichannel OMS + ATP + Order RoutingThe reference OMS for complex omnichannel operations. Manages the complete order lifecycle from capture to delivery with intelligent routing based on cost, lead time, and inventory availability.

SAP S/4HANA — Sales Order Management + aATP — Module: ERP-Integrated OMS + Advanced ATPSAP S/4HANA with the aATP (Advanced Available-to-Promise) module as an OMS integrated with the ERP for companies that don’t require a standalone OMS solution.

Salesforce Order Management — Module: Cloud OMS for B2C RetailSalesforce OMS for B2C retailers with native integration with Salesforce Commerce Cloud and Service Cloud.

4 SO WHAT

Make the OMS the single brain of the order. One capture, one inventory view, routing rules written with the business, and real-time integration with execution.

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